

Flocking Dynamics Simulation

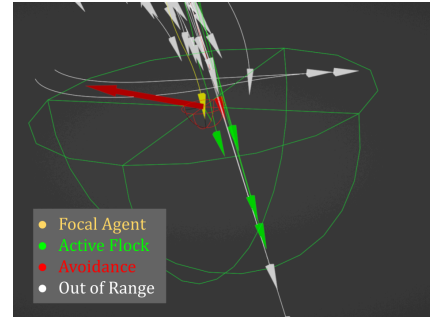
Sole Developer / Mar 2025

Overview:

Simulated flocking behavior using local interaction rules to explore emergent group dynamics, with added behaviors like food-seeking and bounded movement to study external influences on stability and structure.

Tools & Skills:

Blender (Geometry Nodes), Agent-Based Simulation, Procedural Logic, Real-Time Debug Visualization, Behavioral Modeling



Focal agent with visualized behavior zones and nearby influencing agents.

Objective:

Create and experiment with a basic flocking simulation to investigate emergent group behavior through local interaction rules and external stimuli.

Design Highlights:

- Implemented separation, alignment, and cohesion behaviors with adjustable weights to explore their effects on group dynamics.
- Experimented with different neighbor detection methods, including proximity-based and line-of-sight approaches, to study their impact on stability and flock structure.
- Created a visual debug mode to isolate individual agent behavior and visualize contributing influences in real time.
- Introduced a bounded simulation space to maintain localized flocking and simplify observation.
- Integrated a food-seeking behavior where agents detect and compete for visible food sources, consumed by the first to arrive.

Reflections:

Experimenting with rule weighting and detection methods revealed how small changes can significantly affect group behavior and stability. Visual debugging proved valuable for isolating and understanding individual agent responses. Adding boundaries and food-seeking behavior introduced new dynamics and helped highlight the balance between emergent complexity and controllability in agent-based systems. Future improvements could include dynamic obstacle avoidance, energy modeling, predator-prey interaction, or reactive grounding and takeoff behavior.